Unit 9 - Computer Networks
Assignment 16: Managing a LAN Task 9

Introduction

Task description - *You are a junior consultant for T5 Alliance. The work you have done so far for Competition Bikes Ltd has been very well received. As they have no IT support staff of their own, they are interested in outsourcing the ongoing management to your company. First, however, you must prove your knowledge of network management.*

Task 9

(U9:M3) - Report on the business risks of insecure networks and how they can be minimised

Task description - *Explain;*

- *What could happen to an organisations systems and data if it succumbed to a virus attack? What can be done to prevent this?*
- *What could happen to an organisations systems and data if it succumbed to a firewall attack? What can be done to prevent this?*
- *What could happen to an organisation if confidential data was lost or stolen? What can be done to prevent this?*

Introduction

In this report, I will be mentioning the risks that a network is vulnerable to, if it was to have poor implementation of security. These risks include virus attacks, firewall attacks and the loss of confidential data either through being stolen or removed. Basically, I will be mentioning the different drawbacks that could occur when the network is suffering from a risk due to insecurity.

Furthermore, I will also mention what precautions could be taken in order to reduce the said risks from occuring initially. Which would include the essential procedures to reduce the risk of the mentioned impacts the network and the whole organisation would face, essentially how they could reduce and overall prevent these risks from occuring.

Virus attack

Virus attacks within a network have the potential of severely crippling the data and equipment that is connected within the said network. In order for viruses to take effect, they would require

the user interaction in order to attack and spread significantly within the network, through the different workstations (computers). For instance, they would be most commonly attached to emails, which are distributed between users, increasing the changes of the attachment being opened by users which would in turn cause the virus to be executed, attacking and spreading the network, causing many drawbacks.

One drawback that is possible due to the malware attack is the loss of data. Which arguably is a significant impact that a business could face as a consequence of a virus attack, due to the vast importance that the integrity of data has within a network. Essentially, the loss of data would impact on the company's ability to have information, which would then have no reliable source for the staff to carry out work and for customers to access the services.

Furthermore, equipment damage due to virus attacks, is also very possible. This can be due to the fact that hardware may be overloaded with the virus, causing some of the components to reach their stress limits and malfunction as a whole. Meaning that the equipment would need replacement, which would take time to purchase and install, with the addition of staff and other professionals being off duty to carry out this task which could of been avoidable.

With these apparent drawbacks, comes recovery time and costs to the company. Which would again be a significant factor, due to the amount of time and money needed to get the network working at an optimal level. Which again could of been reduced by significant amounts due to the fact this issue is largely preventable.

Due to the time needed to recover the network, it would cause denial of services to the customers and other users that utilise the network to access the desired services. Making the public view of the company as unreliable and inadequate when compared to competitors.

Overall, a virus attack would prove to be fatal not only to the data integrity of the network, but also the risk of the hardware and software malfunctioning within the computer systems. Which would in turn cause the company to waste valuable time and money trying to overcome the issues brought with a virus attack, which should only be a quickly corrected issue.

<u>What can be done to prevent this?</u>

There are multiple precautions set up in place to reduce, and even prevent, the risks of virus attacks within an organisation's network. Which would prove beneficial in it's own respect in preventing virus attacks, but also could cause minor drawbacks with the usage of said prevention techniques. However, these drawbacks are small inconveniences when compared to the drawbacks of a virus attack.

First and foremost is anti-virus software. Basically, this software detects and removes any signs of malicious software within the computer system before it could be executed, which would attack the computer system. It is essentially one of the best ways to insure malware just doesn't

enter the system in the first place. These programs usually work in real-time in the background without the user even knowing things are being monitored, meaning that it doesn't need constant control from a professional.

With the installation of an anti-virus software, it is vital to keep the anti virus software up to date. The reason for this is so the anti-virus software has the adaptability of removing newer viruses and other forms of malware, while also removing older malware in a much more sophisticated manner. Meaning that the software becomes much more efficient with each update.

Additionally, using a good and reputable anti virus would be another benefit due to the fact it is a well funded application that has a wide knowledge of the types of malware that are encountered throughout a majority of companies across the world. Meaning that the company gets constant updates by user feedback or self-found malware and even better ways of dealing with malware.

Additionally, for every company that runs a network, it is vital to have a hardware firewall, typically with a stateful type firewall. Stateful type firewalls is a technique in which it filters the incoming traffic arriving into the network, reducing the chance for a virus to be installed within a workstation. Basically, it can make sure that no suspicious ports or connections get through the network through stateful inspection packet filtering, which essentially tracks each connection travelling across the network.

The firewall is programmed to remove packets that come from an unknown connection, only the packets that come from a known, trusted connection will be allowed through the firewall. They will also maintain a state table that will keep track of all the communication channels, filtering decisions are based not only on user-defined rules, as in packet filtering. Also on context that has been established by prior packets that have passed through the firewall.

However, in order for the above methods to function correctly, it is important to implement a type user awareness training. This is to ensure that every member of staff working for an organisation has up to date knowledge regarding security threats. Which could be done by organising training sessions to ensure the network stays secure, through policies such as the policy regarding CPD for IT professionals.

Essentially, all organisations should have policies in place regarding education and training so the staff and other professionals are able to use the latest software, and are aware of the latest techniques to use when working on the organization's network. While training the staff to use the latest version may cost the company money, but in return it will theoretically enable them to stay more secure.

Overall, while viruses may be a very undesirable form of malware, it is possible to prevent them from occurring initially. Through the means of dedicated software designed to detect and remove, and the implementation of security to reduce the chance of viruses and other malware to pass into the network. Means that the company may spend money for these implementations,

but in turn would reduce the risk of a significantly larger amount of money and time being wasted if a virus attack was to occur.

Firewall hack

A firewall hack is the procedure of an external user either modifying, altering or even removing the said firewall in order to infiltrate and access the data found within the network. Typically, a firewall hack has a wrongful intent of intentionally harming the network and the data found within it, so it is completely undesirable to experience.

The first and foremost is the breach of confidentiality, which can also be known as loss of confidentiality, is also possible due to a firewall hack. It is essentially when information can be viewed by unauthorised users. In a company, the standard confidential information should be customer's personal details. If the information is leaked, it'll show the public how poor their security is and will make it look like they don't care enough to protect their information.

This will essentially also lose trust from the customer and so they'll go to another company. If information so sensitive is stored in a computer, the PC should have additional security such as encryption. If stored physically (which is outdated), it should be kept in a safe in a secret location so only the necessary people can view it.

With the loss of a firewall, it leads the network into further attacks due to the complete loss of security. Which could lead to extremely undesirable consequences of the external offended intentionally causing system harm. Whether it is further hacking into the network and it's valuable data or through a DDoS attack.

A DDoS (distributed denial of service), otherwise known as DoS, attack is another form of intentional harm to the network. Which is essentially the flooding of bandwidth or resources of a network system, but could only be carried out with the lack of a secure firewall. Which would in turn cause significant performance issues would be present with the breach of a network via DDoS. Due to the fact it is basically an external user flooding a network with information to a point where the network would be rendered inaccessible due to the hardware not being able to support the constant amount of significant information.

Additionally, the consequences that may be faced from the firewall being hacked with additional DDoS attacks is the overall loss of services. Essentially, this refers to service from a network being completely deactivated and unusable due to the network being insecure and fails to prevent attack. Meaning, if the company was in a situation like this, their entire network would go down. That would cause the customers and even employees within the company not to have any access to the network and its information. Basically removing the reason why customers and employees would access the company's network and use it's services.

Which would have a major negative impact on the company, specifically with reputation and the overall role of the company, causing the loss of customers. Due to the fact they don't have access to the company website, meaning they'll have to go to an alternative, that would most likely be the competitors. Requiring the company to compensate for the damage and inconvenience.

In order for the company to recover from a situation of this calibre, it would in turn consume valuable time and significant costs in order to get the network back in working order. Similar to a company experiencing a virus attack, it would consume significant amounts of time and money in order to get the network and data back in working order. With the additional cost of legal issues of the situation.

Overall, the result of a firewall being hacked could bring in many potential risks to an organisation. Whether it is the exposure of valuable data to the public and any other unauthorised access or vulnerability to a wider array of more harmful attacks causing complete loss of services. Hence why it's important to make sure that the company's network and security is up to date and in working order, so they don't end up in such a critical situation.

What can be done to prevent this?

In order to keep the company's network and security up to date and in working order, there are systems which can help prevent the hacking of a firewall through means of complex security which either include the ability to detect an interception or the ability to create a false firewall or even a false network for the perpetrator.

Firstly, could be the implementation of an IDS, or intrusion detection systems. Essentially, an intrusion detection system is used to detect unauthorised entries, alerting an administrator to respond. An IDS inspects all inbound and outbound network activity and identifies suspicious patterns that may indicate a network or system attack from someone attempting to compromise a system.

Additionally, a better quality firewall, most likely a proxy firewall, could be used in order to lure away an attacker from the firewall in the first place, through the means of disallowing direct connection. Typically, proxy firewalls are very secure, this does however come at the expense of speed and functionality.

Proxy firewalls are secure because unlike other types of firewall, data packets don't pass through a proxy. Instead, the proxy acts as a mirror and makes a new network connection based on the request. This prevents direct connections meaning it is harder for attackers to discover the location of the network. It does this by receiving the request, then it looks over it for suspicious information before allowing that data to reach the protected network.

The benefits of using proxy firewalls are that it is a very secure firewall type by looking at information packets on the application layer, also breaking the connection between trusted and untrusted systems.

However, proxy firewalls can only support a limited number of applications, traffic performance may be negatively affected and the denial of untrusted connections could cause functionality issues. Although, the drawbacks are miniscule compared to the consequences of being hacked.

In order for a proxy firewall to function is through acceptable configuration of the firewall. Meaning that the IT professional needs to have a good understanding of firewalls before the configuration of the firewall. However, with a satisfactory configuration, makes it extremely difficult to intercept the network firewall.

Just as importantly, regular checks with the status of the firewall by a professional is vital in order to check any faults, errors or even signals of interception of the firewall. Meaning that the IT professional would know what to carry out given the accurate estimation of the current circumstances.

Essentially, It is vital for a professional to monitor the firewall regularly. So it is important to have policies in place regarding computer monitoring for all IT employees. Monitoring refers to watching the activity of the network and firewall to ensure there aren't signs of suspicious activity. Also, arbitrary monitoring should take place to ensure that the network stays secure and no that any error or suspicious activity would be noticed early.

One feature firewalls have to ease the monitoring of the firewall is through alerts. Which is essentially the application sending a message to the professional currently monitoring the firewall. In which the message will indicate that there is an issue, giving information on what type of issue it is. Where the professional will manage the issue accordingly.

Additionally, if an interception was attempted, an IT professional could check the log files in order to check what activity was carried out, and whether the cause of the activity is traceable. Giving the company a chance of finding the perpetrator and carrying out appropriate legal action.

To improve the chances of a hacker not being able to infiltrate a network, is through the installation of multiple firewalls. Which essentially add extra levels of security which slows down the process of the security hack. Either making the perpetrator give up on their efforts or giving the professionals enough time to trace where the attack is originating from, giving the ability of legal action.

ISPs also have levels of blockage of the network at a Network Level, which help aid the security of the network with the additional firewalls set inplace. Meaning that the ISP can block certain

individuals outside the ISP connection accessing certain areas within the network, even the entire network.

Also, a honeypot system could be installed as another method of misleading the attacker into a false attack. Essentially, This is a system where a server would be set up in the screened subnet or demilitarised zone in an attempt to lure attackers to it.

This would be set up separately from the actual server, and will hold false information. This will mislead the attacker into believing they have attacked the organisation's true server. In order to make this server attractive to attackers, the organisation would leave some ports open, to ease the attack of the honeypot.

However, to help make the honeypot more realistic, the server would need contain some security software. Usually, this software will be easy enough to get through, but will still reassure the attacker they have found the correct server. While the attacker is trying to gain access to the false server, the organisation can monitor what the attacker does. This is so that they can trace the interceptor and apprehend the suspect and prevent future attacks to the real server, improving overall security.

Overall, reducing the probability of the firewall and network being targeted and attacked by an interceptor it extremely important for the company and its data integrity. With methods such as an improved firewall, the stacking of firewalls or even misleading the attacker as method of preventing said consequences. Some professionals may even use detailed logs to gain the identity of the attacker and notify the local authorities.

Loss of confidential data

The maintenance and integrity of confidential data is vital for a company and its public image. Which is important since a public image is the first way a customer is introduced to a company. Essentially, the better the company's reputation, the more customers they will get.

As mentioned before, is again the loss of services. If the network of the organisation is insecure and fails to prevent attack, they will lose that service. It could be due to a technical fault, in which the company would then request help from the technicians, or it could be a hacking situation or even a security breach.

Regardless of what the issue is, it has a major negative impact on the company. Cause a mass loss of customers, as they won't have access to the company's service meaning they'll have to go to an alternative such as competitors.

Additionally, the leak of confidential data would mean that the valuable data of users, employees and even the company would be leaked to the public. Which gives the public the ability to access the confidential information with ease. With the competitors also being able to

obtain the leaked data with ease, giving competitors the advantage over the company due to the extra information. This would in turn cause a significant cost in the recovery of data. Due to the fact that the organisation would have to spend significant amounts of money and time through the recovery of data. Whether it is the costs of retrieving the data from the public or a ransom and storing it in a much more secure network.

However, there are significant legal consequences, which may refer to the consequences when a company breaks one of the computing laws. To elaborate, if the company was to break the Data Protection Act and expose their information to the public, not only will it damage the company's reputation and trust, but it will personally damage the victims of this act. Making it possible for the victims of this act to face fines or jail, due to the violation of the said data protection act. In turn giving the company serious reputational damage with the customers, a vast majority of employees and other partnerships the company has.

Therefore causing brand image damage, essentially giving the company the image of being unreliable and reckless with the storage and safekeeping of confidential data. Also, with the company branded as a criminal company, causes significant damage which would be impossible to recover from.

Overall, the loss of confidential data brings multiple consequences not only to the integrity of company data within a network, but the necessary private information needed in order to keep the organisation functioning as a business and the customer's vital information, such as credit card numbers and home addresses. Which bring in multiple legal and reputable issues to a company.

What can be done to prevent this?

In terms of maintaining confidential data, there are multiple ways in which this can be secured and decrease the chance of the data being leaked into the public. Which can cause a variety of issues, as seen previously.

Encryption is one of the most effective ways of securing data. Essentially, the encryption of data is the process of converting the data into an unreadable format. Which the user would then need a secret key or password in order to decrypt it. So, when sending data, it will be encrypted before it is sent and then decrypted when it is received by the other user. Even if information does find its way towards another person who isn't supposed to view it.

Access can be denied through a password or special passphrase, this way whoever wants to view the document needs that encryption key to view and modify. Furthermore this kind of technology enables files to be transparently encrypted to protect confidential data from attackers with physical access to the computer that might decide to break in.

For example, there are forms of encryption on many programs such as text editors, windows folders, USB, Tapes, CD's and other databases storages. These kind of encryptions serve as a barrier and, although it can still be cracked, it is harder to do so due to the fact it is a longer process overall.

It is also very necessary to have clear policies regarding backup. Essentially, a backup is taken at the end of each day to ensure all work completed that day can not be lost, and is stored safely. Usually at the end of each month or week, all backups are checked to ensure they are being taken correctly. Backups are essential in any organisation to ensure no important files are ever lost due to multiple reasons.

Also, it can be restored even with the risk of information being stolen. However, there needs to be maintenance on electronic equipment, due to the fact that hardware could malfunction or any other disaster would would damage the equipment. To overcome this, the files can be backed up to places such as the cloud or simply another server and many of the times they can be free. So, if a disaster occured and all the information that was stored in the servers are lost, all the backups would be safe as there far away from this scenario. Also, backups are capable of being automatically set as a specific time.

As mentioned previously, firewalls are designed to prevent unauthorised access to a computer or network. You can implement a firewall in both hardware and software, or a combination of both. A firewall will monitor data packets coming in and out of the network it is protecting and will enforce the company's network security policy.

It filters out the packets that look suspicious and do not meet the specified security criteria. Most organisations use firewalls to protect their network from the Internet. However it is just as important for a business to have a software firewall in place to monitor connections on a workstation or server, this can act as an extra line of defence as monitors in real-time.

With both of these firewalls in place it means intruders can't snoop around on the computers and steal valuable information. The same thing can be said to virus protection as a good antivirus is required to stop malware entering the system. Not only does this slow down the work process but can also damage the software preinstalled in the system, which means businesses to have to reinvest on all this again, basically increasing the costs.

An anti-virus could also be used. virus protection as a good antivirus is required to stop malware entering the system. Not only does this slow down the work process but can also damage the software preinstalled in the system, which means businesses to have to reinvest on all this again, basically increasing the costs.

Through the use of access control lists, or ACL, the network could have a better level of data security. ACLs are essentially lines of text that the firewall will apply to each packet of data it

receives, these lines of text provide specific information defining what packets can be accepted, and what packets must be denied depending on the user.

The main advantage of using a packet filtering firewall is that they are very flexible, you can easily customise the firewall and allow it to work with many different protocols and applications. Also, they are not application-dependant and they are capable of working at high speeds because they do not carry out extensive processing on the data packets. Although, due to the small number of variables used in access control decisions, they are susceptible to security breaches caused by improper configurations and they also cannot prevent attacks that employ application-specific vulnerabilities.

Finally, restricted access could be secure method of keeping data secure within a company, since it is executed via directories, making the computer network robust. This can prevent any unauthorised access to functions such as the disk drivers, servers and other accounts. This means even if a user manages to gain access through physical use, they still wouldn't be able to access applications and other files within the software. So in a business a lower level employees might not get access to the higher level documents so even if someone did manage to get onto their account.

Lastly, VLANs could be implemented within the network to significantly increase the chances of the data from leaving the network initially. Essentially, a VLAN is a virtual LAN, which is a much more isolated variant of a LAN. Giving the ability of data being stored securely, without any possible method of accessing unless it is through the hardware in which it is stored in, which is usually restricted with an encrypted password.

Overall, the maintenance of confidential data should be top priority for an organisation with the integration of a network. Not only due to the fact that this data is essential for a business point of view, but also in a legal and ethical point of view. Due to the fact that this data contains sensitive information which would put the individuals that this data applies to at a severe risk when exposed.

Conclusion

In conclusion, there are multiple risks faced when dealing with insecure networks, especially for the company and its overall image towards the public. Since there is the risk of facing legal action for the data protection act offenses. Also, the poor implementation of security can cost the company a significant amount of time, workforce and money wasted on issues which could easily be avoided by a suitable implementation of security.

Overall, keeping the security of a network within a company is essential in order to reduce the risks which the company is exposed to by having the implementation of a network. Meaning that having the optimal amount of security, reduces the waste of unnecessary money, time and workforce on dealing with preventable issues. Inturn increasing the survivability of the company.